GSE 518
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Suggested Solutions to Problem Set 5

1. From our results from problem set 3 question 4 we know that if $X \sim N(1,4)$ and $Z = (X-1)/2$, then $Z \sim N(0,1)$. Using this fact we have $\Pr(X<1) = \Pr\left(\frac{X-1}{2} < \frac{1-1}{2}\right) = \Pr(Z<0) = 0.5$, and $\Pr(X \geq 1.5) = 1 - \Pr(X \leq 1.5) = 1 - \Pr\left(\frac{X-1}{2} < \frac{1.5-1}{2}\right) = 1 - \Pr(Z<0.25) = 1 - 0.5987 = 0.4013$.

2. In this case we may think of an individual showing up for his/her flight as a Bernoulli trial with 0.90 probability of "success" (i.e., showing up). We want to model the number of persons that actually show up for the flight out of the 23 reservations the company takes. Let $X =$"number of individuals that show up for the flight". Therefore, we can model X as a Binomial distribution with $p = 0.90$ and $n = 23$, or $Binomial(23, 0.90)$. The probability that the airline can accommodate everybody is given by $\Pr(X \leq 20)$, which equals

$$\begin{aligned}
\Pr(X \leq 20) &= 1 - \Pr(X > 20) = 1 - \Pr(X=21) - \Pr(X=22) - \Pr(X=23) \\
&= 1 - \left(\frac{23!}{(23-21)!21!}\right)(0.9)^{21}(0.1)^{2} - \left(\frac{23!}{(23-22)!22!}\right)(0.9)^{22}(0.1)^{1} \\
&\quad - \left(\frac{23!}{(23-23)!23!}\right)(0.9)^{23}(0.1)^{0} \\
&= 1 - \left(\frac{23 \times 22}{2}\right)(0.9)^{21}(0.1)^{2} - (23)(0.9)^{22}(0.1)^{1} - (0.9)^{23} \\
&= 0.4080
\end{aligned}$$

Hence, there is a 40.80% chance that the airline is able to accommodate everybody.

3. If X follows a Poisson distribution then its pmf is given by $\Pr(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$ for $x = 0, 1, \ldots$. Given that $\Pr(X=1) = \Pr(X=2)$, we have that

$$\lambda e^{-\lambda} = \frac{\lambda^2 e^{-\lambda}}{2!}$$

which implies $\lambda = 2$. Therefore, we have that $\Pr(X=3) = \lambda^3 e^{-\lambda}/3! = 2^3 e^{-2}/6 = 0.18044$.

4.
(a) From CB we can see that the mgf of Z when $Z \sim Gamma(\alpha, \beta)$ is given by $M_Z(t) = \left(\frac{1}{1-\beta t}\right)^{\alpha}$ for $t < \frac{1}{\beta}$; and the mgf of Y when $Y \sim Chi\text{-}Squared(p)$, is given by $M_Y(t) =$

$\left(\frac{1}{1-2t}\right)^{p/2}$ for $t < \frac{1}{2}$. Therefore, if $X \sim Gamma(p/2, 2)$ then its mgf is $M_X(t) = \left(\frac{1}{1-2t}\right)^{p/2}$, which is the mgf of a $Chi\text{-}Squared(p)$ random variable and equals the mgf of Y. Therefore, by Theorem 2.3.11 we know that since X and Y have $M_X(t) = M_Y(t)$ for all t in an interval around zero, then $X \sim Y$. In other words, a $Gamma(\alpha, \beta)$ with $\alpha = p/2$ and $\beta = 2$ equals a $Chi\text{-}Squared(p)$.

(b) Here we obtain the mgf of $Y = X^2$ when $X \sim N(0,1)$ and show it equals the mgf of a $Chi\text{-}Squared(1)$. From part (a) we know that the mgf of a random variable Z which follows a $Chi\text{-}Squared(1)$ is given by $M_Z(t) = \frac{1}{1-2t}$. The mgf of $Y = X^2$ is given by

$$\begin{aligned} M_Y(t) &= E\left(e^{tY}\right) = E(e^{tX^2}) = \int_{-\infty}^{\infty} e^{tx^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{(-\frac{1}{2}x^2 + tx^2)} dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-2t)x^2}{2}} dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2/(1-2t)}} dx \\ &= \frac{1}{\sqrt{(1-2t)}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{1/(1-2t)}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2/(1-2t)}} dx = \frac{1}{(1-2t)^{1/2}} \end{aligned}$$

where in the last equality we used the fact that the term inside the integral is the pdf of a normal $N(0, 1/(1-2t))$ and therefore it integrates to 1. As mentioned in class, this is a very common trick used in statistics to work with integrals. Therefore, $M_Y(t) = \frac{1}{(1-2t)^{1/2}}$, which is equal to the mgf of a $Chi\text{-}Squared(1)$. Therefore, by Theorem 2.3.11, $Y \sim Chi\text{-}Squared(1)$.

5. The version of Chebyshev's inequality we saw in class states that for any random variable Y with mean μ and variance σ^2 we have that, for any $k > 0$, $\Pr(|Y - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$. In this case we want to calculate the bound $\Pr(|X - \mu| \geq k\sigma)$ for a random variable X and with $k = 2$ using Chebyshev's inequality. Note that $k\sigma > 0$, and therefore Chebyshev's inequality implies that $\Pr(|X - \mu| \geq k\sigma) \leq \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2} = \frac{1}{4}$. Therefore, by Chebyshev's inequality we have that $\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{4}$.

We now calculate the actual probability. Given the information we have that $\mu = E(X) = (-1)\Pr(X = -1) + (0)\Pr(X = 0) + (1)\Pr(X = 1) = -\frac{1}{8} + \frac{1}{8} = 0$; $E(X^2) = (-1)^2 \Pr(X = -1) + (0)^2 \Pr(X = 0) + (1)^2 \Pr(X = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$; so $\sigma^2 = Var(X) = E(X^2) - [E(X)]^2 = \frac{1}{4}$. Therefore, in this case we have that $\Pr(|X - \mu| \geq k\sigma) = \Pr(|X| \geq 2(1/2)) = \Pr(|X| \geq 1) = \Pr(X = -1) + \Pr(X = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$. Hence, in this example the actual probability $\Pr(|X - \mu| \geq k\sigma)$ equals Chebyshev's upper bound.